

Hepatitis A Virus (HAV) Antibody, Total With Reflex to IgM

Test Details

METHODOLOGY

Immunochemiluminometric assay (ICMA); (See individual tests for methodologies)

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This test is used to aid in the diagnosis of active Hepatitis A (HAV) infection and differentiate between active and previous infection/vaccination.

HAV Antibody Testing Interpretation Chart

HAV Total Antibody	HAV IgM	Comments
Negative	Not done	No evidence of vaccination or previous infection; Susceptible to Hepatitis A infection
Positive	Not done	Consistent with recent or remote Hepatitis A infection or antibody response to HAV vaccination
Positive	Negative	Consistent with resolved Hepatitis A infection or antibody response to HAV vaccination
Positive	Positive	Consistent with active Hepatitis A infection

Special Instructions

If reflex test is performed, additional charges/CPT code(s) may apply.

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

This assay has not been FDA cleared or approved for the screening of blood or plasma donors. Assay performance characteristics have not been established for immunocompromised or immunosuppressed patients, cord blood or patients less than 2 years of age.

The performance characteristics of this assay have not been determined for use in cadaveric specimens.

Custom Additional Information

HAV is a picornavirus primarily transmitted via the fecal-oral route. HAV replicates in the liver and is shed in high concentrations in feces from 2-3 weeks before to 1 week after the onset of clinical illness. IgM antibody develops within a week of symptom onset, peaks around three months, and is usually no longer detectable after six months. Many cases of hepatitis A are subclinical, particularly in children. Antibody produced in response to HAV infection (anti-HAV) persists for life and confers protection against reinfection. The presence of IgM antibody to HAV is diagnostic of acute HAV infection. A positive test for total anti-HAV indicates immunity to HAV infection but does not differentiate current from previous HAV infection. Although usually not sensitive enough to detect the low level of protective antibody after vaccination, anti-HAV tests also might be positive after hepatitis A vaccination.

Specimen Requirements

Specimen

Serum **or** EDTA plasma

Volume

2 mL

Minimum Volume

0.8 mL

Container

Red-top tube, gel-barrier tube, **or** lavender-top (EDTA) tube, serum transfer tube, plasma transfer tube

Collection Instructions

If tube other than a gel-barrier tube is used, transfer separated serum/plasma to a labeled plastic transport tube. Do **not** freeze gel-barrier tube (pour off serum first).

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Room temperature

Causes for Rejection

Heat-inactivated specimens; cord blood; cadaveric specimens; or body fluid other than serum or EDTA plasma; gross hemolysis; excessive lipmia; improper labeling

References

Miller JM, Binnicker MJ, Campbell S, et al. A guide to utilization of the microbiology laboratory for diagnosis of infectious diseases: 2018 update by the Infectious Disease Society of America and the American Society for Microbiology. *Clin Infect Dis*. 2018 Aug 31;67(6):813-816. PubMed 30169655 (<http://www.ncbi.nlm.nih.gov/pubmed/30169655>)

Nelson NP, Weng MK, Hofmeister MG, et al. Prevention of Hepatitis A Virus Infection in the United States: Recommendations of the Advisory

Committee on Immunization Practices, 2020. *MMWR Recomm Rep*. 2020 Jul 3;69(5):1-38. PubMed 32614811 (<http://www.ncbi.nlm.nih.gov/pubmed/32614811>)